Step1: Install required tools

```
charan@LT-NSPIRA:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [717 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [868 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [157 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [195 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [110 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [220 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [389 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [72.5 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [122 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [62.7 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [586 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [602 kB]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [122 kB]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [20.0 kB]
Get:21 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [3728 B]
Get:22 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [125 kB]
Get:23 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [21.0 kB]
Get:24 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [4168 B]
Get:25 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [3136 B]
Get:26 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Fetched 5060 kB in 30s (169 kB/s)
147 packages can be upgraded. Run 'apt list --upgradable' to see them.
charan@LT-NSPIRA:~$ sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd
netcat-openbsd is already the newest version (1.234-1).
netcat-openbsd set to manually installed.
Upgrading:
  iproute2  libpcre2-8-0  libprotobuf-lite32t64  zlib1g
```

```
Installing dependencies:
  gnome-themes-extra-data  libhwloc15          libprotoc32t64            libsm6            libx264-165
  ibverbs-providers        libibverbs1         libqt6core5compat6        libsmi2t64        libx265-215
  libaacs0                 libice6             libqt6core6t64            libsnappy1v5      libxcb-cursor0
  libabsl20260107          libinput-bin        libqt6dbus6               libsoxr0          libxcb-icccm4
  libavcodec62             libinput10          libqt6gui6                libspandsp2t64    libxcb-image0
  libavformat62            libiperf0           libqt6multimedia6         libspeexdsp1      libxcb-keysyms1
  libavutil60              libjsoncpp26        libqt6network6            libsrt1.5-gnutls  libxcb-render-util0
  libb2-1                  liblua5.4-0         libqt6opengl6             libssh-4          libxcb-shape0
  libbcg729-0              libmaxminddb0       libqt6printsupport6       libsvtav1enc2     libxcb-util1
  libbdplus0               libmbedcrypto16     libqt6qml6                libswresample6    libxcb-xinput0
  libbluray3               libmd4c0            libqt6qmlmeta6            libswscale9       libxcb-xkb1
  libcares2                libminizip1t64      libqt6qmlmodels6          libtbb12          libxkbcommon-x11-0
  libchromaprint1          libmtdev1t64        libqt6qmlworkerscript6    libtbbbind-2-5    libxnvctrl0
  libcjson1                libnghttp3-9        libqt6quick6              libtbbmalloc2     libxvidcore4
  libcodec2-1.2            libnl-3-200         libqt6svg6                libts0t64         libze1
  libdav1d7                libnl-genl-3-200    libqt6waylandclient6      libudfread3       libzmq5
  libdouble-conversion3    libnl-route-3-200   libqt6waylandcompositor6  libva-x11-2       libzvbi-common
  libdvdnav4               libnorm1t64         libqt6widgets6            libvdpau1         libzvbi0t64
  libdvdread8t64           libopencore-amrnb0  libqt6wlshellintegration6 libvorbisfile3    ocl-icd-libopencl1
  libevdev2                libopengl0          librabbitmq4              libvpl2           qt6-gtk-platformtheme
  libfalcosecurity0t64     libopenjp2-7        librav1e0.8               libwacom-common   qt6-qpa-plugins
  libgme0                  libopenmpt0t64      libre2-11                 libwacom9         qt6-svg-plugins
  libgrpc++1.51t64         libpcap0.8t64       librist4                  libwireshark-data qt6-translations-l10n
  libgrpc29t64             libpcre2-16-0       libsbc1                   libwireshark19    qt6-wayland
  libgsm1                  libpgm-5.3-0t64     libsctp1                  libwiretap16      wireshark-common
  libhwloc-plugins         libprotobuf32t64    libshine3                 libwsutil17

Suggested packages:
  libcuda1           libdvdcss2               lksctp-tools          geoip-database              wireshark-doc
  libnvcuvid1        libhwloc-contrib-plugins snmp-mibs-downloader  geoip-database-extra        opencl-icd
  libnvidia-encode1  mmdb-bin                 libwacom-bin          libjs-leaflet
  libbluray-bdj      qt6-qmltooling-plugins   geoipupdate           libjs-leaflet.markercluster
```

Step2: Configure host1, router and host2

```
charan@LT-NSPIRA:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
charan@LT-NSPIRA:~$ sudo ip netns exec host1 ip link set veth-h1 up
charan@LT-NSPIRA:~$ sudo ip netns exec host1 ip link set lo up
charan@LT-NSPIRA:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
charan@LT-NSPIRA:~$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
charan@LT-NSPIRA:~$ sudo ip netns exec router ip link set veth-r1 up
charan@LT-NSPIRA:~$ sudo ip netns exec router ip link set veth-r2 up
charan@LT-NSPIRA:~$ sudo ip netns exec router ip link set lo up
charan@LT-NSPIRA:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
charan@LT-NSPIRA:~$ sudo ip netns exec host2 ip link set veth-h2 up
charan@LT-NSPIRA:~$ sudo ip netns exec host2 ip link set lo up
charan@LT-NSPIRA:~$
```

Step3: Enable routing

```
charan@LT-NSPIRA:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
charan@LT-NSPIRA:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
charan@LT-NSPIRA:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
charan@LT-NSPIRA:~$
```

Step4: Test Connection

```
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=2.18 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.058 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.059 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2034ms
rtt min/avg/max/mdev = 0.058/0.766/2.182/1.001 ms
charan@LT-NSPIRA:~$
```

Step5: Add network delay

```
charan@LT-NSPIRA:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
charan@LT-NSPIRA:~$
```

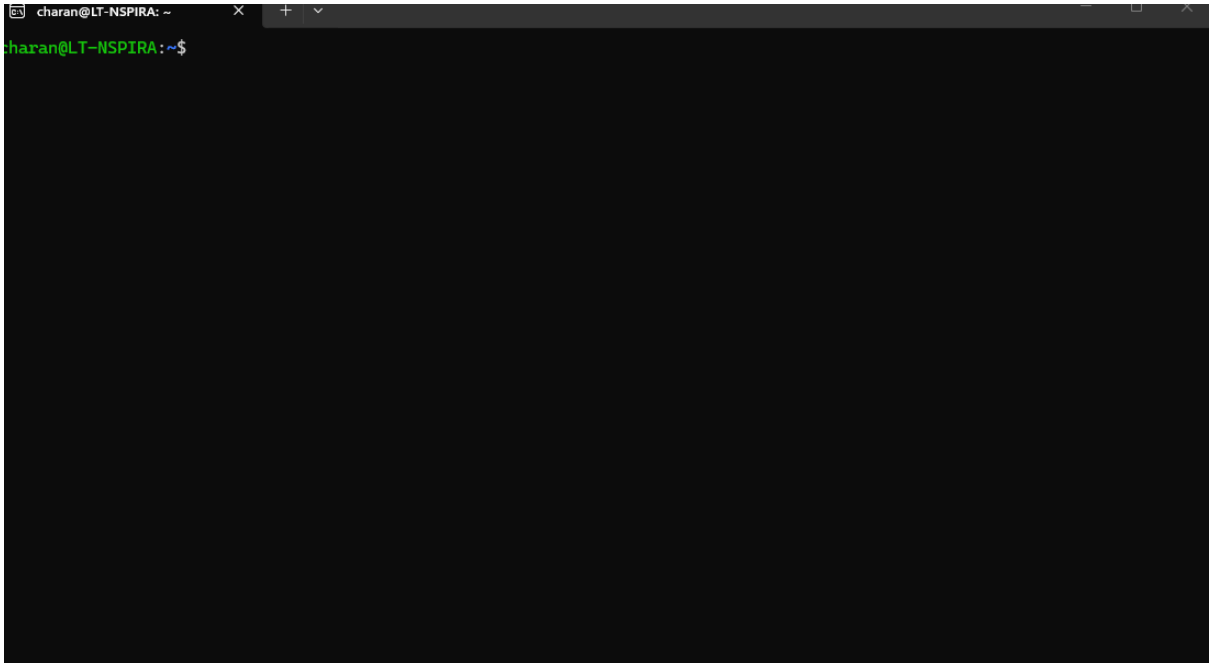
Step6: Verify Network delay

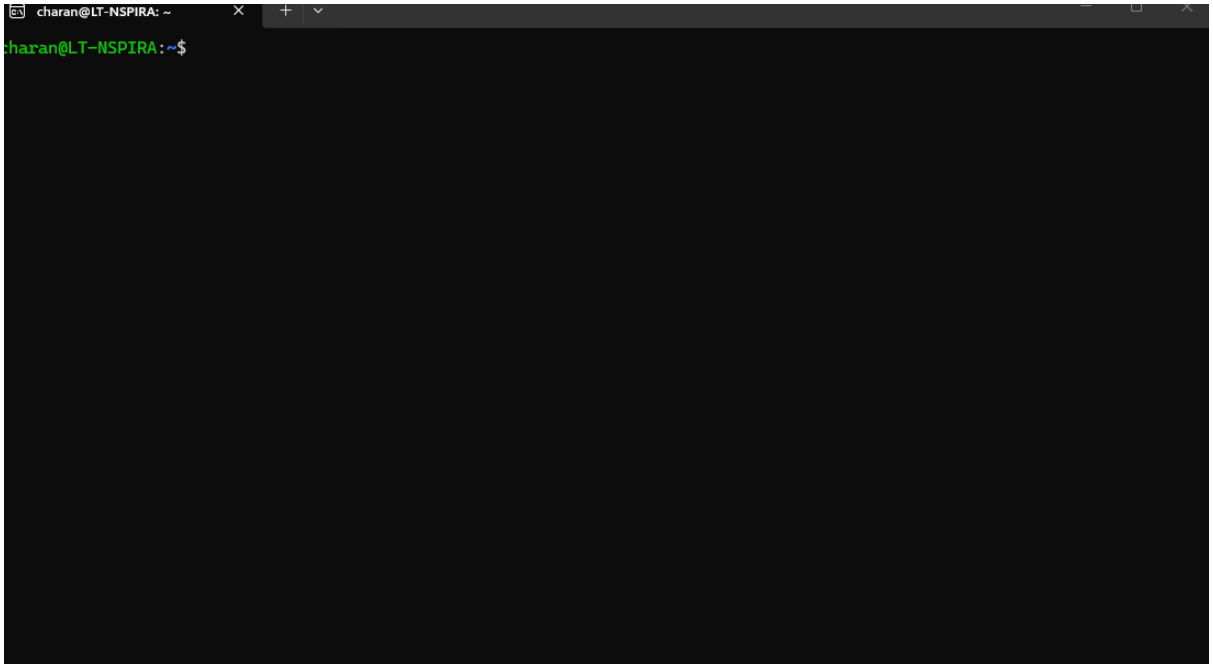
```
charan@LT-NSPIRA:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
charan@LT-NSPIRA:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 9 limit 1000 delay 20ms loss 1% seed 10742690013224494828
charan@LT-NSPIRA:~$
```

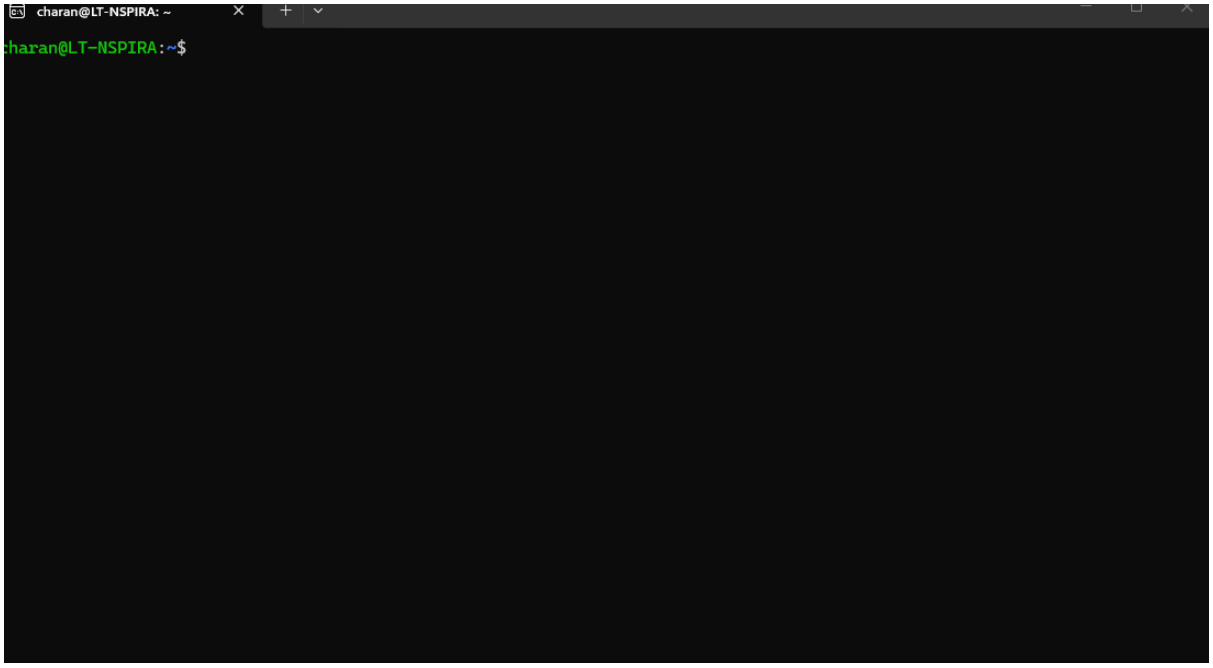
Step7: Open 4 ubuntu Terminals

```
charan@LT-NSPIRA: ~
charan@LT-NSPIRA:~$ sudo ip netns add host1
Cannot create namespace file "/run/netns/host1": File exists
charan@LT-NSPIRA:~$ sudo ip link set veth-h1 netns host1
Cannot find device "veth-h1"
charan@LT-NSPIRA:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
charan@LT-NSPIRA:~$ sudo ip netns exec host1 ip link set veth-h1 up
charan@LT-NSPIRA:~$ sudo ip netns exec host1 ip link set lo up
charan@LT-NSPIRA:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
charan@LT-NSPIRA:~$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
charan@LT-NSPIRA:~$ sudo ip netns exec router ip link set veth-r1 up
charan@LT-NSPIRA:~$ sudo ip netns exec router ip link set veth-r2 up
charan@LT-NSPIRA:~$ sudo ip netns exec router ip link set lo up
charan@LT-NSPIRA:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
charan@LT-NSPIRA:~$ sudo ip netns exec host2 ip link set veth-h2 up
charan@LT-NSPIRA:~$ sudo ip netns exec host2 ip link set lo up
charan@LT-NSPIRA:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
charan@LT-NSPIRA:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
charan@LT-NSPIRA:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
charan@LT-NSPIRA:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=2.18 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.058 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.059 ms

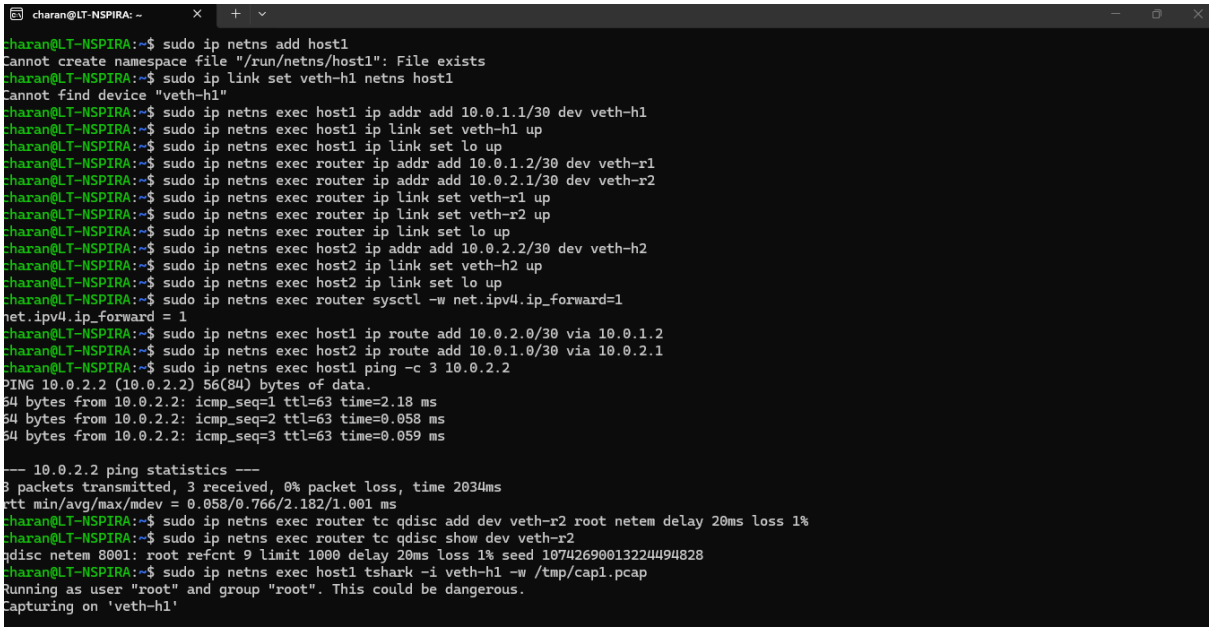
--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2034ms
rtt min/avg/max/mdev = 0.058/0.766/2.182/1.001 ms
charan@LT-NSPIRA:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
charan@LT-NSPIRA:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 9 limit 1000 delay 20ms loss 1% seed 10742690013224494828
charan@LT-NSPIRA:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
```

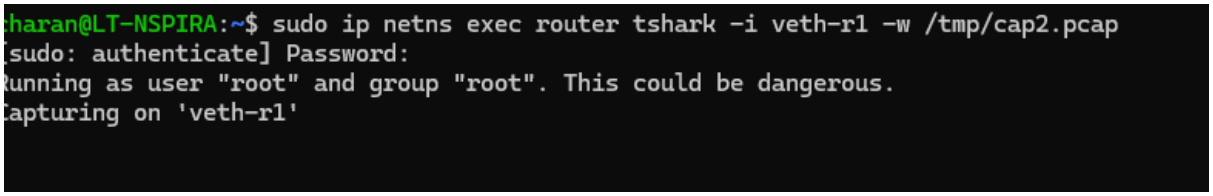

charan@LT-NSPIRA: ~
charan@LT-NSPIRA:~$


charan@LT-NSPIRA: ~
charan@LT-NSPIRA:~$



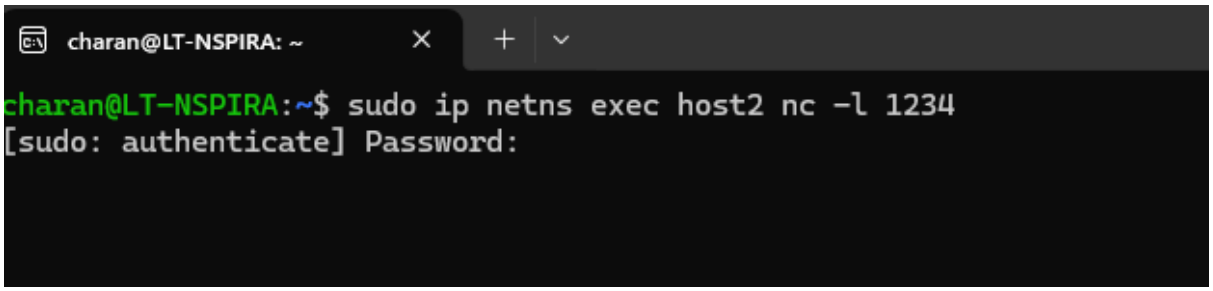
Step8: Start capturing



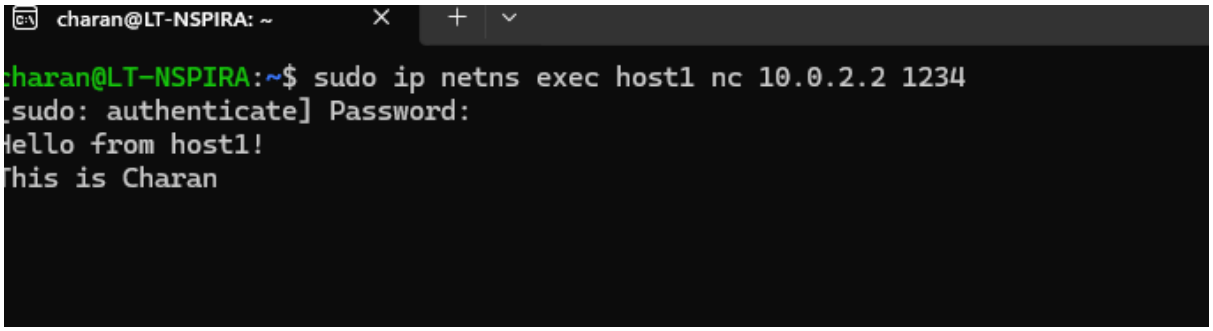




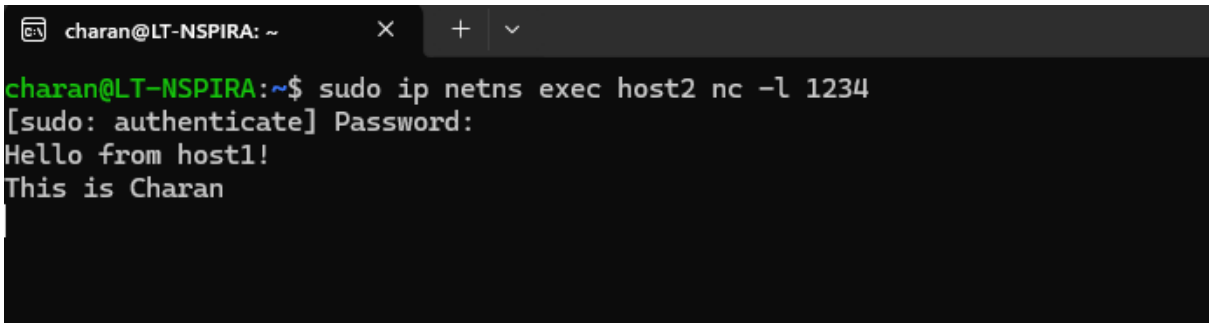
Step9: Start receiver



Step10: Send message in 5th terminal



Messages should be received in 4th terminal



Step11: Go to the terminal and stop packet capturing

```
charan@LT-NSPIRA:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
17
^C
charan@LT-NSPIRA:~$
```

```
charan@LT-NSPIRA:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
17
^C
charan@LT-NSPIRA:~$
charan@LT-NSPIRA:~$
```

```
charan@LT-NSPIRA:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
15 ^C
charan@LT-NSPIRA:~$
```

```
charan@LT-NSPIRA:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
Hello from host1!
This is Charan
^C
charan@LT-NSPIRA:~$
```

```
charan@LT-NSPIRA:~$ sudo ip netns exec host1 nc 10.0.2.2 1234
[sudo: authenticate] Password:
Hello from host1!
This is Charan
^C
charan@LT-NSPIRA:~$
charan@LT-NSPIRA:~$
```

Step12: Fix file permissions and check capture files in terminal 4

```
charan@LT-NSPIRA:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
Hello from host1!
This is Charan
^C
charan@LT-NSPIRA:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
charan@LT-NSPIRA:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 1.9K Nov 12 05:23 /tmp/cap1.pcap
-rw-r--r-- 1 root root 1.9K Nov 12 05:23 /tmp/cap2.pcap
-rw-r--r-- 1 root root 1.8K Nov 12 05:23 /tmp/cap3.pcap
charan@LT-NSPIRA:~$
```

Step13: Check TTl

```
charan@LT-NSPIRA:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
charan@LT-NSPIRA:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 1.9K Nov 12 05:23 /tmp/cap1.pcap
-rw-r--r-- 1 root root 1.9K Nov 12 05:23 /tmp/cap2.pcap
-rw-r--r-- 1 root root 1.8K Nov 12 05:23 /tmp/cap3.pcap
charan@LT-NSPIRA:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      3 63
      4 64
charan@LT-NSPIRA:~$ tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      3 63
      4 64
charan@LT-NSPIRA:~$ tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      4 63
      3 64
charan@LT-NSPIRA:~$
```

Step14: Read your message from router capture

```
charan@LT-NSPIRA:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e tcp.payload 2>/dev/null | xxd | head -20
00000000: 0a0a 0a34 3836 3536 6336 6336 6632 3036  ...48656c6c6f206
00000010: 3637 3236 6636 6432 3036 3836 6637 3337  6726f6d20686f737
00000020: 3433 3132 3130 610a 0a35 3436 3836 3937  431210a..5468697
00000030: 3332 3036 3937 3332 3034 3336 3836 3137  3206973204368617
00000040: 3236 3136 6530 610a 0a                   2616e0a..
charan@LT-NSPIRA:~$
```

Step15: Check Delay

```
00000040: 3236 3136 6530 610a 0a                   2616e0a..
charan@LT-NSPIRA:~$ tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
48.517555938
charan@LT-NSPIRA:~$ tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
48.517411899
charan@LT-NSPIRA:~$ tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
32.155144664
charan@LT-NSPIRA:~$
```

Step16: Check Transmissions

```
32.155144664
charan@LT-NSPIRA:~$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
charan@LT-NSPIRA:~$ tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
charan@LT-NSPIRA:~$ tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
charan@LT-NSPIRA:~$
```

Step17: Check Rtt

```
charan@LT-NSPIRA:~$ sudo ip netns exec host1 mtr --report 10.0.2.2
Start: 2026-11-12T05:37:11+0000
HOST: LT-NSPIRA                   Loss%   Snt   Last   Avg  Best  Wrst StDev
  1.|-- 10.0.1.2                   0.0%    10    0.1   0.4   0.1   2.8   0.8
  2.|-- 10.0.2.2                   0.0%    10   20.3  20.7  20.3  22.0   0.5
charan@LT-NSPIRA:~$
```

Step18: Open Wireshark

```
10.0.1.1       10.0.2.2       64
charan@LT-NSPIRA:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234 && tcp.len > 0" -T fields -e tcp.payload | while read x; do echo "$x" | xxd -r -p;
 done
Hello from host1!
This is Charan
charan@LT-NSPIRA:~$ tshark -r /tmp/cap2.pcap -q -z io,stat,0

===================================
| IO Statistics                   |
|                                 |
| Duration: 100.6 secs            |
| Interval: 100.6 secs            |
|                                 |
| Col  1: Frames and bytes        |
|---------------------------------|
|               |1                |
| Interval      | Frames | Bytes  |
|---------------------------------|
|   0.0 <> 100.6 |     17 |   987 |
===================================
charan@LT-NSPIRA:~$ tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" | wc -l
0
charan@LT-NSPIRA:~$ sudo ip netns del host1
charan@LT-NSPIRA:~$ sudo ip netns del router
charan@LT-NSPIRA:~$ sudo ip netns del host2
charan@LT-NSPIRA:~$ sudo ip netns list
charan@LT-NSPIRA:~$
```